double that amount of time, two days, to break a 65-bit key, and four days to break a 66-bit key. Breaking a 128-bit key would take 2^{64} times longer than a day, or fifty million billion years. By that time, I might even be pardoned.

In my communications with journalists, I used 4096- and 8192-bit keys. This meant that absent major innovations in computing technology or a fundamental redefining of the principles by which numbers are factored, not even all of the NSA's cryptanalysts using all of the world's computing power put together would be able to get into my drive. For this reason, encryption is the single best hope for fighting surveillance of any kind. If all of our data, including our communications, were enciphered in this fashion, from end to end (from the sender end to the recipient end), then no government—no entity conceivable under our current knowledge of physics, for that matter—would be able to understand them. A government could still intercept and collect the signals, but it would be intercepting and collecting pure noise. Encrypting our communications would essentially delete them from the memories of every entity we deal with. It would effectively withdraw permission from those to whom it was never granted to begin with.

Any government hoping to access encrypted communications has only two options: it can either go after the keymasters or go after the keys. For the former, they can pressure device manufacturers into intentionally selling products that perform faulty encryption, or mislead international standards organizations into accepting flawed encryption algorithms that contain secret access points known as "back doors." For the latter, they can launch targeted attacks against the endpoints of the communications, the hardware and software that perform the process of encryption. Often, that means exploiting a vulnerability that they weren't responsible for creating but merely found, and using it to hack you and steal your keys—a technique pioneered by criminals but today embraced by major state powers, even though it means knowingly preserving devastating holes in the cybersecurity of critical international infrastructure.

The best means we have for keeping our keys safe is called "zero knowledge," a method that ensures that any data you try to store externally—say, for instance, on a company's cloud platform—is encrypted by an algorithm running on your device before it is uploaded, and the key is never shared. In the zero knowledge scheme, the keys are in the users' hands—and only in the users' hands. No company, no agency, no enemy can touch them.

My key to the NSA's secrets went beyond zero knowledge: it was a zero-knowledge key consisting of multiple zero-knowledge keys.